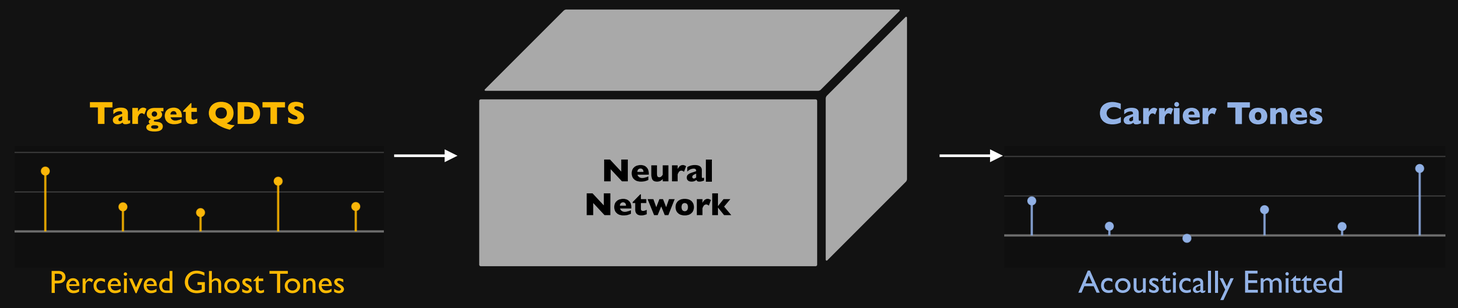
Neural Quadratic Difference Tones Spectra (QDTS)

A Max device that synthesizes harmonic phantom tones — perceived by the listener but never acoustically emitted — by computing the carrier complex needed to evoke a desired target QDTS in the auditory system.

there is established literature on how a given carrier complex produces a specific target QDTS; however, computing the reverse — finding the carrier complex needed to evoke a desired target — is a hard mathematical problem.

This device solves it in real time using a neural network trained to approximate that inverse mapping.



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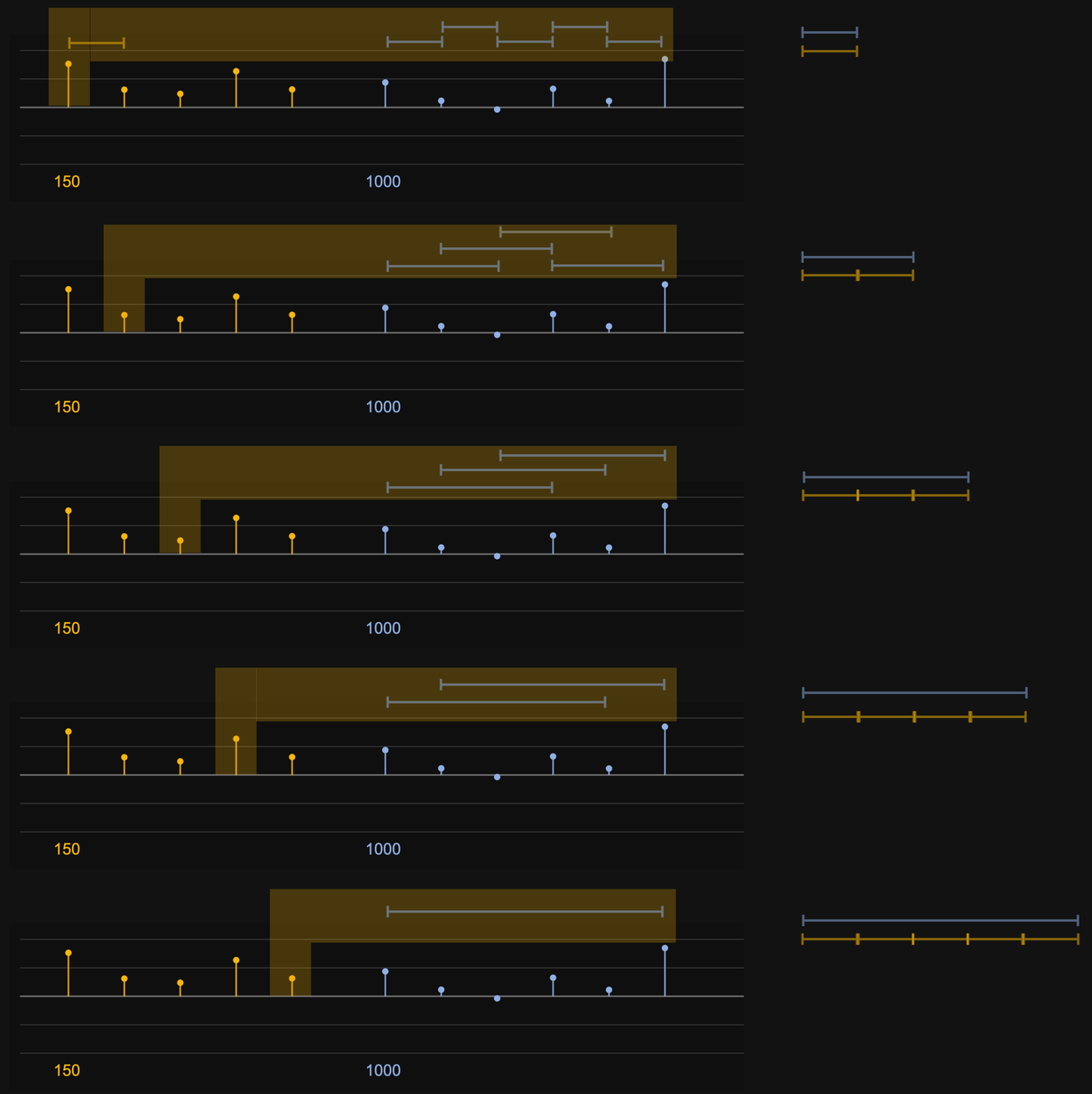
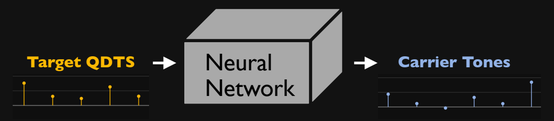
The differences between the carrier complex tones (shown in cyan) result in a harmonic series of phantom tones perceived by the listener (shown in orange).

Note: To perceive *N* harmonics in the phantom tones, *N+1* carrier complex tones are required



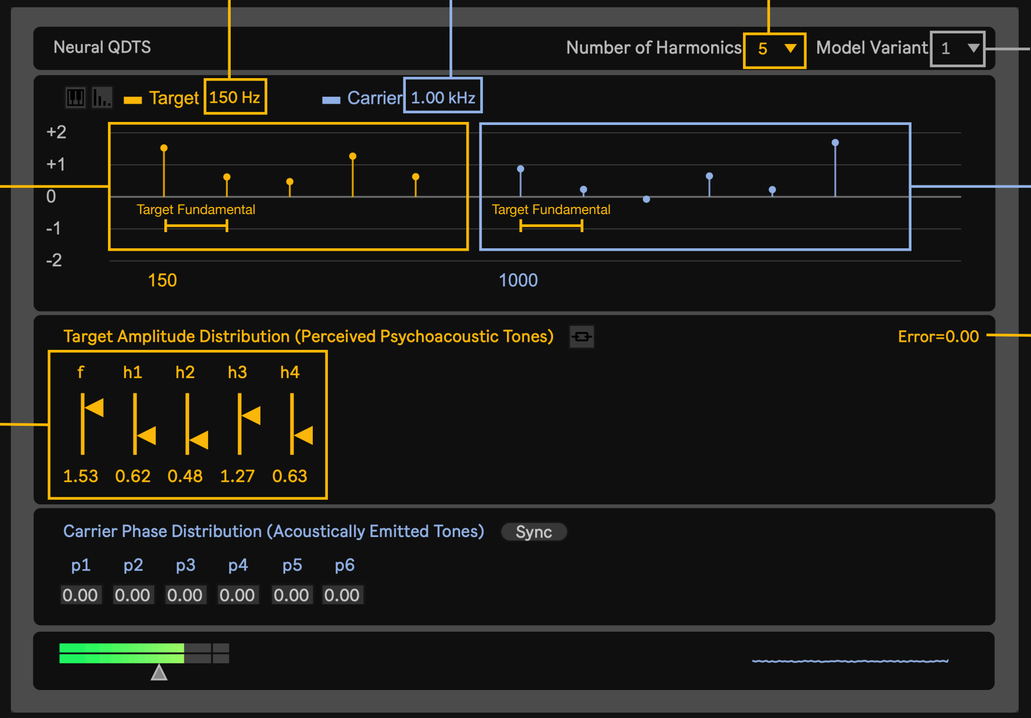
Interface



Sets the fundamental frequency of the target QDTS

Sets the lowest frequency of the carrier complex

Sets the number of harmonics in the target QDTS



Selects between different trained neural network models for computing the carrier complex

Target QDTS: The harmonic series of phantom tones to be perceived by the listener, starting at frequency F

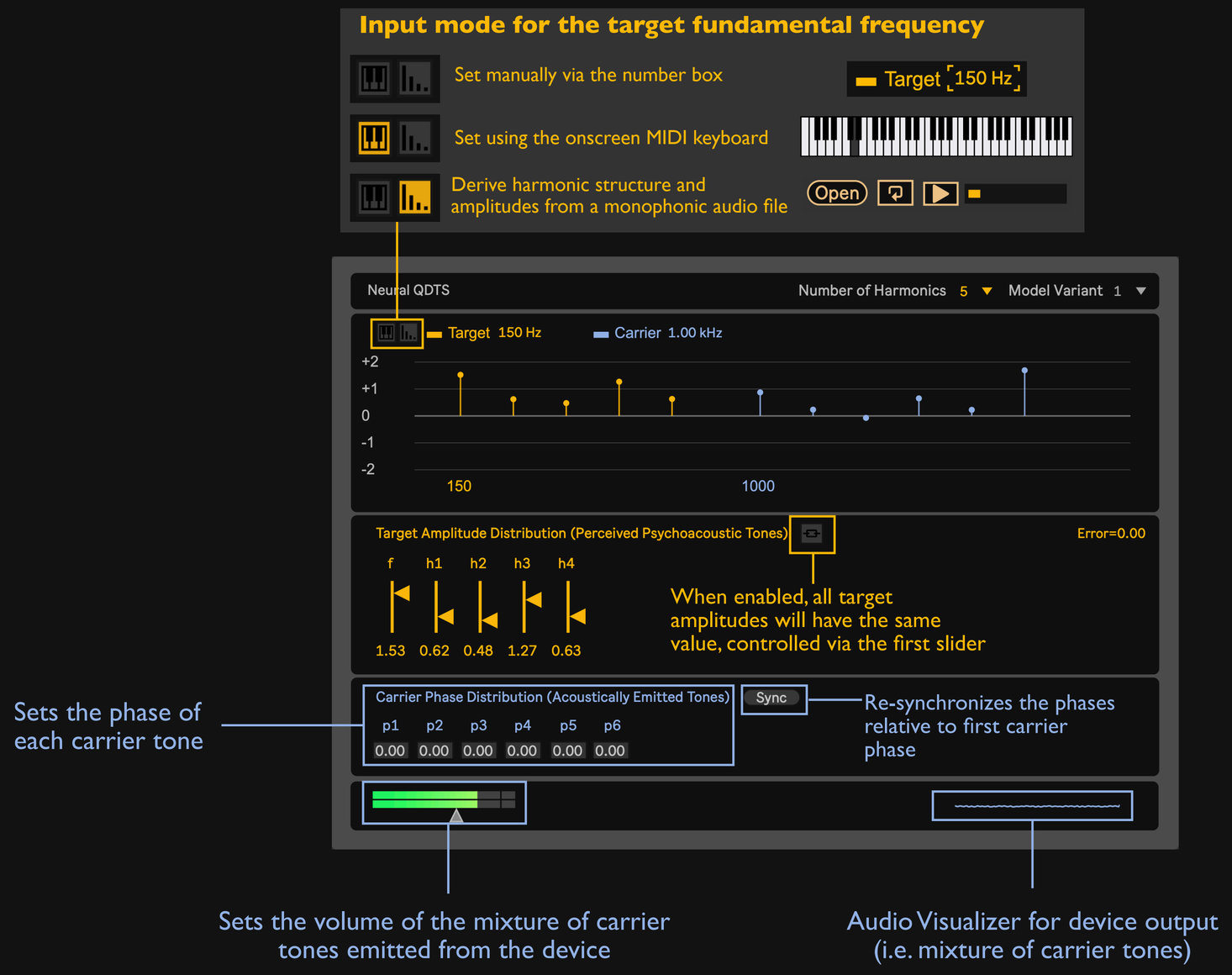
Carrier Tones: The acoustic signal emitted by the speakers, computed by the neural network to evoke the target QDTS.

Indicates how well the predicted carrier tones can result in the perception of the specified phantom tone configuration

Sets the amplitude of each harmonic in the target QDTS

The neural network computes the carrier amplitudes needed to evoke the specified target QDTS

Interface


Input mode for the target fundamental frequency
Set manually via the number box
Target 150 Hz
Set using the onscreen MIDI keyboard
Derive harmonic structure and amplitudes from a monophonic audio file
Open
Neural QDTS
Number of Harmonics 5
Model Variant 1
Target 150 Hz
Carrier 1.00 kHz
+2
+1
0
-1
-2
150
1000
Target Amplitude Distribution (Perceived Psychoacoustic Tones)
Error=0.00
f h1 h2 h3 h4
1.53 0.62 0.48 1.27 0.63
When enabled, all target amplitudes will have the same value, controlled via the first slider
Carrier Phase Distribution (Acoustically Emitted Tones)
Sync
p1 p2 p3 p4 p5 p6
0.00 0.00 0.00 0.00 0.00 0.00
Re-synchronizes the phases relative to first carrier phase
Sets the phase of each carrier tone
Sets the volume of the mixture of carrier tones emitted from the device
Audio Visualizer for device output (i.e. mixture of carrier tones)